 Name:_____

1 The General Chain Rule

THEOREM 5.3 (THE CHAIN RULE) Suppose $X \subseteq \mathbf{R}^m$ and $T \subseteq \mathbf{R}^n$ are open and $\mathbf{f}: X \to \mathbf{R}^p$ and $\mathbf{x}: T \to \mathbf{R}^m$ are defined so that range $\mathbf{x} \subseteq X$. If $\mathbf{x}$ is differentiable at $\mathbf{t}_0 \in T$ and $\mathbf{f}$ is differentiable at $\mathbf{x}_0 = \mathbf{x}(\mathbf{t}_0)$, then the composite $\mathbf{f} \circ \mathbf{x}$ is differentiable at $\mathbf{t}_0$, and we have

$$D(\mathbf{f} \circ \mathbf{x})(\mathbf{t}_0) = D\mathbf{f}(\mathbf{x}_0) D\mathbf{x}(\mathbf{t}_0).$$

In matrix notation, this reads

$$\begin{pmatrix} \frac{\partial f_1}{\partial t_1} & \frac{\partial f_1}{\partial t_2} & \cdots & \frac{\partial f_1}{\partial t_n} \\ \frac{\partial f_2}{\partial t_1} & \frac{\partial f_2}{\partial t_2} & \cdots & \frac{\partial f_2}{\partial t_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_p}{\partial t_1} & \frac{\partial f_p}{\partial t_2} & \cdots & \frac{\partial f_p}{\partial t_n} \end{pmatrix} = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_m} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_m} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_p}{\partial x_1} & \frac{\partial f_p}{\partial x_2} & \cdots & \frac{\partial f_p}{\partial x_m} \end{pmatrix} \begin{pmatrix} \frac{\partial x_1}{\partial t_1} & \frac{\partial x_1}{\partial t_2} & \cdots & \frac{\partial x_1}{\partial t_n} \\ \frac{\partial x_2}{\partial t_1} & \frac{\partial x_2}{\partial t_2} & \cdots & \frac{\partial x_2}{\partial t_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial x_m}{\partial t_1} & \frac{\partial x_m}{\partial t_2} & \cdots & \frac{\partial x_m}{\partial t_n} \end{pmatrix},$$

where each $\frac{\partial x_i}{\partial t_j}$ is evaluated at $\mathbf{t}_0$ and each $\frac{\partial f_k}{\partial x_\ell}$ is evaluated at $\mathbf{x}(\mathbf{t}_0)$.

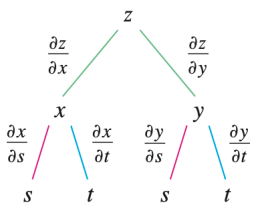
Example 1.1. Suppose $z = f(x,y)$ with $x = x(s,t)$ and $y = y(s,t)$. Then

$$\begin{aligned} \begin{pmatrix} \frac{\partial z}{\partial s} \\ \frac{\partial z}{\partial t} \end{pmatrix} &= \begin{pmatrix} \frac{\partial z}{\partial x} & \frac{\partial z}{\partial y} \end{pmatrix} \begin{pmatrix} \frac{\partial x}{\partial s} & \frac{\partial x}{\partial t} \\ \frac{\partial y}{\partial s} & \frac{\partial y}{\partial t} \end{pmatrix} \\ &= \begin{pmatrix} \frac{\partial z}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s} \\ \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t} \end{pmatrix} \end{aligned}$$

which implies

$$\begin{aligned} \frac{\partial z}{\partial s} &= \frac{\partial z}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s} \\ \frac{\partial z}{\partial t} &= \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t}. \end{aligned}$$

Note that in this case we have three types of variables: s and t are independent variables, x and y are intermediate variables, and z is the dependent variable. To remember these formulas, it may be helpful to draw the following tree diagram:



1. If $z = e^x \sin(y)$, where $x = st^2$ and $y = s^2t$, find $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t}$.

2. Write out the Chain Rule for the case where $w = f(x,y,z,t)$ and $x = x(u,v), y = y(u,v), z = z(u,v)$, and $t = t(u,v)$.

3. If $u = x^4y + y^2z^3$, where $x = rse^t, y = rs^2e^{-t}$, and $z = r^2s\sin(t)$, find the value of $\frac{\partial u}{\partial s}$ when $r = 2, s = 1, t = 0$.

4. If $g(s,t) = f(s^2 - t^2, t^2 - s^2)$ and f is differentiable, show that g satisfies the equation

$$t\frac{\partial g}{\partial s} + s\frac{\partial g}{\partial t} = 0.$$

5. If $z = f(x,y)$ has continuous second-order partial derivatives and $x = r^2 + s^2$ and $y = 2rs$, find

   (a) $\frac{\partial z}{\partial r}$

   (b) $\frac{\partial^2 z}{\partial r^2}$